

Risk-Based Prioritization of Heated-Pavement Deployment - Public Excerpt

Public-safe technical excerpt derived from the original project output.

Source: 2025 Gwangjin-gu Big-Data Analysis Competition report

Objective

Prioritize Gwangjin-gu road segments for heated-pavement installation using winter road-risk indicators and model probabilities.

Model result

Random Forest ROC-AUC: approximately 0.888 in the submitted analysis.

Decision output

The five highest-priority roads were selected by predicted probability and then reviewed against the real road environment.

Limitations

The project noted temporal mismatch across public-data sources, spatial approximation error, limited winter-road coverage, and class imbalance.